



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
30-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

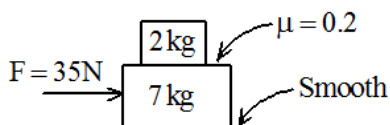
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

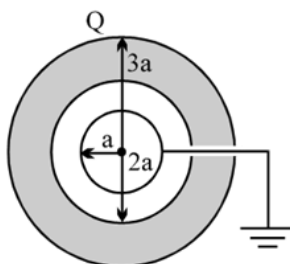
1. A Force of magnitude 35 Newtons is applied on a two-block system as shown in the figure. If the coefficient of friction between the two blocks is 0.2 while the horizontal surface is smooth then choose the **CORRECT** statement(s). (Take : $g = 10 \text{ m/s}^2$)



- (A) The frictional force acting between the two blocks is 4 Newtons
- (B) The frictional force acting between the two blocks is 2 Newtons
- (C) The blocks move without slipping with common acceleration
- (D) There is slipping with the lower block accelerating faster than the upper one.

Space for Rough Work

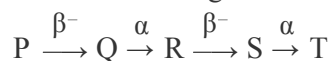
2. In the figure shown, both the shells are conducting. The outer shell is thick & has a total charge of Q , inner shell is thin & is earthed.



- (A) Charge on inner shell is $(-2Q/5)$
 (B) Charge on inner shell is $(-3Q/5)$
 (C) Total charge on outer shell remain Q
 (D) Charge on outer surface of outer shell is $(3Q/5)$

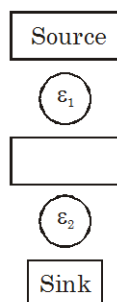
Space for Rough Work

3. A nucleus undergoes a series of decay according to the scheme



atomic no and mass number of T are 57 & 139.

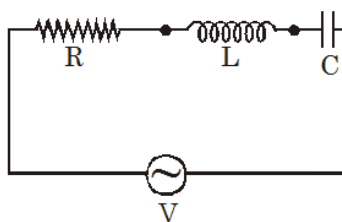
- (A) Atomic number of P is 59
(B) Mass number of Q is 143
(C) Atomic number of R is 58
(D) Mass number of S is 143
4. Two engines operate in series between a heat sources & sink as shown in figure. Heat source supplies 600 J to engine ϵ_1 and has efficiency 40%. Engine ϵ_2 has efficiency 25%. Choose the **CORRECT** option/s :



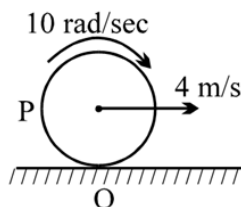
- (A) Work done by engine ϵ_1 is 360J
(B) Work done by engine ϵ_2 is 90J
(C) Efficiency of complete system is 55%
(D) Heat rejected to sink is 270 J

Space for Rough Work

5. An AC source is in series with R-L-C as shown in figure. Choose the **CORRECT** option/s:
(given : $R = 8\Omega$, $X_C = 6\Omega$, $X_L = 12\Omega$ & $V = 50 \sin(\omega t + 37^\circ)$ in volts)



- (A) Power produced at $t = \pi/2\omega$ is 200 watt
(B) Power produced at $t = \pi/2\omega$ is 250 watt
(C) Magnitude of voltage across inductor at $t = \pi/2\omega$ is 60 volt
(D) Magnitude of voltage across inductor at $t = \pi/2\omega$ is zero
6. A disc of radius 20 cm and mass 1 kg is rolling with slipping on a flat horizontal surface. At a certain instant, the velocity of its center is 4 m/s and its angular velocity is 10 rad/s. The lowest contact point is O.



- (A) The velocity of point O is 2 m/s
(B) The distance of instantaneous center of rotation(ICOR) from the point O is 0.2 m
(C) Angular momentum of disc about ICOR is $1.8 \text{ kgm}^2/\text{sec}$
(D) Angular momentum of disc about center of mass of disc is $0.2 \text{ kgm}^2/\text{sec}$

Space for Rough Work

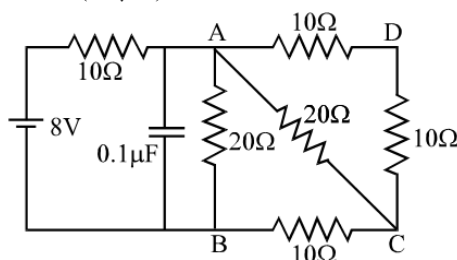
SECTION-II : (Maximum Marks: 24)

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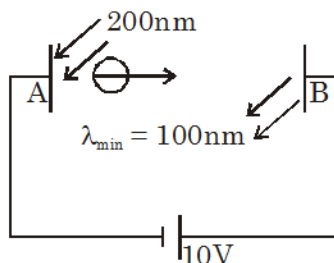
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A capacitor of capacitance $0.1 \mu\text{F}$ is connected to a battery of emf 8V as shown in the fig. Under steady state condition find charge on capacitor (in μC).

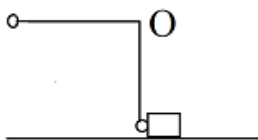


2. In the figure shown electromagnetic radiations of wavelength 200 nm are incident on the metallic plate A. The photo electrons are accelerated by a potential difference 10V . These electrons strike another metal plate B from which electromagnetic radiations are emitted. The minimum wavelength of the emitted photons is 100 nm . Find the work function of the metal 'A' (in eV). Use $hc = 12400 \text{ eV}\text{\AA}$



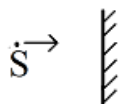
Space for Rough Work

3. A U tube has 2 tubes of radius 1 mm and 1.5 mm. What will be the difference in water level (in mm) in two limbs? (Assume Contact angle $\theta = 0^\circ$, surface tension $= 7.5 \times 10^{-2}$ N/m, $g = 10$ m/s²).
4. A ball of 0.01 kg is suspended from a fixed point O by means of a light and inextensible string of length 10 m. Initially string is released (rest) from horizontal position as shown in figure. At the lowermost point of its path, it hits a stationary small block of mass 1.205 kg lying on a rough horizontal surface. After striking the block, the ball rebounds and comes to rest when the string makes an angle 30° with the horizontal. Due to impact the block moves through a distance of 0.5 m on the horizontal surface. If the coefficient of friction between the block and the surface is k , find value of $10k$. (Take : $g = 10$ m/s²)



Space for Rough Work

5. A transverse travelling wave represented by a equation: $y = (1 \text{ cm}) \sin[2\pi x - 20\pi t]$; where x is in meters and t is in seconds, is travelling on a string of mass per unit length 0.01 Kg/m . Find the kinetic energy (in mJ) of a portion of the string of length equal to one wavelength.
6. A source S having a detector D , moving towards a wall with a certain velocity detects 9 beats/sec. On doubling the velocity of the source the detector D detects 20 beats/sec. The original frequency of the sound emitted by the source is 5°C Hz. Find C . (speed of sound in air is 330 m/s)



Space for Rough Work

SECTION-III : (Maximum Marks: 18)

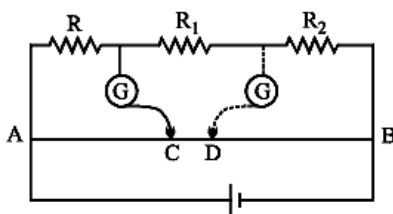
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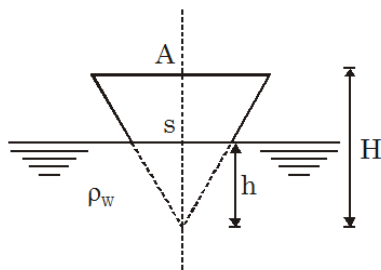
1. The diagram shows a modified meter bridge, which is used for measuring two unknown resistance at the same time. When the first galvanometer is used for obtaining the balance point, it is found at point C. Now the first galvanometer is removed and the second galvanometer is used, which gives balance point at D. Using the details given in the diagram, find out the value of $10 \times \frac{R_2}{R_1}$. (Given: $AB = L$; $AC = L/4$ and $AD = 2L/3$)



2. A projectile is projected with speed v m/s such that magnitude of displacement of projectile in 2nd and 7th sec is same and equals to $5 \times \sqrt{217}$. Find value of $v/10$. (Take : $g = 10 \text{ m/s}^2$)

Space for Rough Work

3. A heater supplies constant power 80 watt to an object which have steady state temperature equals to 40°C . When heater is switched off, its initial rate of cooling is $8^{\circ}\text{C}/\text{sec}$ and rate of cooling of object is $5^{\circ}\text{C}/\text{sec}$ when its temperature is 28°C . If temperature of surrounding is $T^{\circ}\text{C}$, find value of T .
4. A solid cone made of a material of relative density $\left(s = \frac{27}{64}\right)$ and height $H = 4\text{m}$ floats with its apex downward in water. The time period of vertical oscillation if it is slightly displaced from the equilibrium position is T , find value of T . (Assume $\pi^2 = g$)



Space for Rough Work

5. A biconvex lens of radius of curvature R is made up of a variable refractive index $\mu = 2 \left(1 + \frac{|y|}{d} \right)$. Assume $2d \ll R$ where d is radius of aperture. A point object is placed at $R = 7.5$ m from the lens on the principal axis. If the spread of image lies over a span of n meter, find value of n . (Here y is perpendicular distance of lens material from principle axis.)
6. A two slit Young's interference experiment is done with monochromatic light of wavelength $6000\text{\AA}$. These slits are 2 mm apart. The fringes are observed on a screen placed 10 cm away from the slits. A transparent plate of thickness 0.5 mm is placed in front of one of the slits. It is found that the interference pattern shifts by 5 mm. If the refractive index of the transparent plate is μ , find value of 5μ .

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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1. The wave function of two different orbitals X and Y in Be^{+3} ion are given by -

$$\psi_X = \frac{1}{9\sqrt{3}} \left(\frac{1}{4\pi} \right)^{1/2} \left(\frac{z}{a_0} \right)^{3/2} (6 - 6\sigma + \sigma^2) e^{-\sigma/2}$$

$$\psi_Y = \frac{1}{9\sqrt{6}} \left(\frac{3}{4\pi} \right)^{1/2} \left(\frac{z}{a_0} \right)^{3/2} (4 - \sigma)\sigma e^{-\sigma/2} \cos\theta$$

$$\sigma = \frac{2zr}{na_0}$$

where $a_0 = 1^{\text{st}}$ Bohr radius of H-atom, z = Charge number of nucleus, r = distance from nucleus

From this we can conclude :

(A) Total number of nodal surface is same for both orbitals

(B) The angular nodal surface of Y orbital will occur at $\theta = \frac{\pi}{2}$

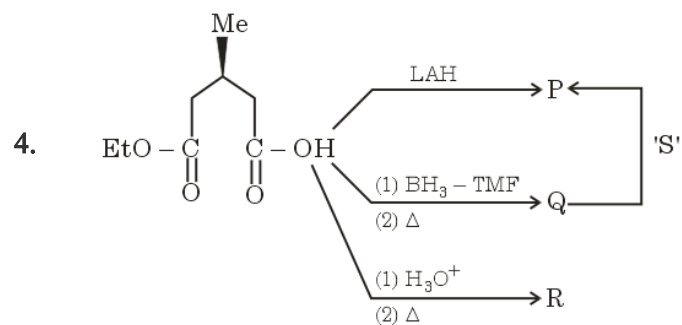
(C) Energy of electron in Y-orbital is greater than in X-orbital

(D) Average distance of X orbital from nucleus is greater than Y orbital

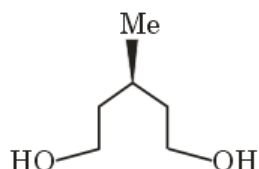
Space for Rough Work

2. Select correct statement(s).
- (A) $(\text{CH}_3)_3\text{COH}$ is less acidic as compared to $(\text{CH}_3)_3\text{SiOH}$
- (B) $(\text{CH}_3)_2\text{N}(\text{SiH}_3)$ is more basic than $(\text{CH}_3)\text{N}(\text{SiH}_3)_2$
- (C) NO_2 more easily dimerizes as compared to NO
- (D) BCl_3 is better L.P. acceptor (Lewis acid) as compared to BF_3 & AlCl_3
3. Two gases X and Y, one being the dimer of other are at equilibrium. Increase of volume at constant temperature or increase of temperature at constant pressure favours the formation of more Y. The reaction could not be represented by $[Q > 0]$
- (A) $2\text{X} \rightleftharpoons \text{Y} + Q \text{ calories}$
- (B) $2\text{X} \rightleftharpoons \text{Y} - Q \text{ calories}$
- (C) $2\text{Y} \rightleftharpoons \text{X} - Q \text{ calories}$
- (D) $2\text{Y} \rightleftharpoons \text{X} + Q \text{ calories}$
-

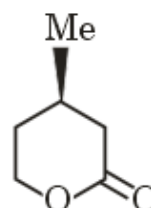
Space for Rough Work



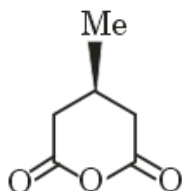
(A) "P" product is =



(B) "Q" product is =

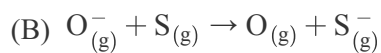
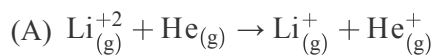


(C) "R" product is =



(D) "S" reagent is = LiAlH_4

5. Select process which is/are exothermic ($\Delta H = -ve$) :

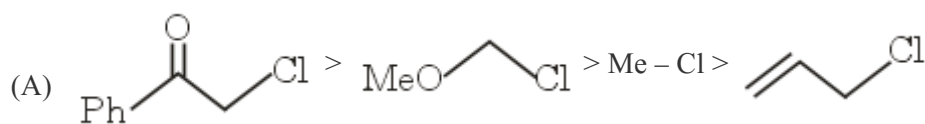


(C) Conversion of tetrachloridocobaltate(II) ion to hexaaquacobalt(II) ion in aqueous solution.

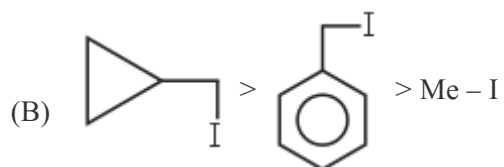
(D) dioxygenium ion + e^{-} $\rightarrow$ molecular oxygen

Space for Rough Work

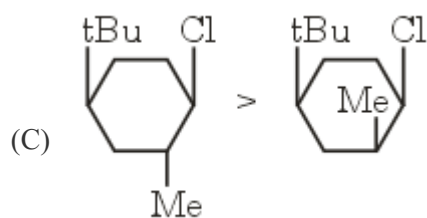
6. Identify correct reactivity order for various alkyl halides



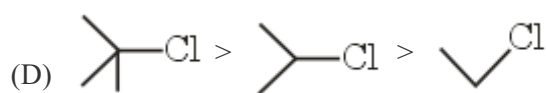
reactivity order with NaI / acetone



rate of hydrolysis ($\text{S}_\text{N}1$)



rate of E_2 reaction with alc.KOH



rate of E_1 reaction

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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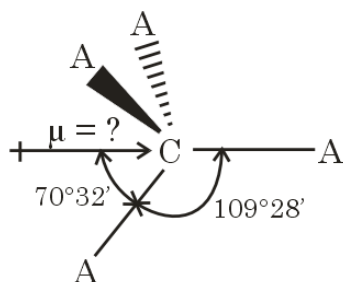
1. The emf of a cell formed by dipping two electrodes of metal (M) in two solutions of its salt in which the concentration of metal ion (M^{n+}) is 0.1 M and 0.001 M was observed to be 0.03 volt at 25° C. What is the valency of the metal ion if the two half cells are connected using salt bridge.

$$\text{given } 2.303 \frac{RT}{F} = 0.06$$

Space for Rough Work

2. In C-A bond (where C is carbon and A is an univalent atom) atom A has 0.4 fraction of +ve charge due to polarity of bond with $1\text{\AA}$ internuclear distance. A perfectly tetrahedral molecule CA_4 , has the symmetrical tetrahedral structure with the C-atom at the centre and the four identical valences making an angle of $109^\circ 28'$ with each other. The centre of symmetry is at the C-atom. The $-\text{CA}_3$ group has a regular pyramidal configuration with C-atom at its apex and three A's at the base. The resultant moment of the three C-A bonds will be directed along a line from the center of the base towards the apex and opposite to the direction of the fourth C-A bond moment. Using bond moment for $\overrightarrow{\text{C}-\text{A}}$ bond, find the resultant dipole moment of the regular pyramidal $-\text{CA}_3$ group.

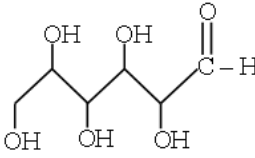
(in Debye unit) (given that $\cos 109^\circ 28' = -\frac{1}{3}$, $\cos 70^\circ 32' = \frac{1}{3}$, $\cos 180^\circ = -1$)



3. 50 ml of solution of 0.2 M NaOH is mixed with 50 ml of 0.1 M $\text{La}(\text{NO}_3)_3$ (aq.) solution. How many millimoles of La^{3+} ions will be precipitated.

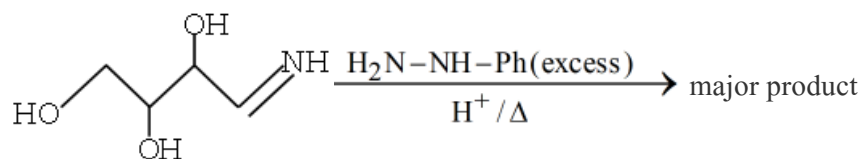
$[\text{K}_{\text{sp}} \text{ of } \text{La}(\text{OH})_3 = 10^{-16}]$

Space for Rough Work

4. Identify total number of aldohexose with  structure which give optically inactive product on reaction with HNO_3 ?

5. Consider complex salt $[\text{Fe}(\text{H}_2\text{O})_6]_3 [\text{Fe}(\text{CN})_6]_2$
Find $(x + y + z)$
 x = average number of unpaired electrons on each iron ion.
 y = total number of 3d-orbitals involved in hybridisation for the formation of one molecule of above complex
 z = average oxidation number of iron

6. Identify gm molecular mass of major product obtain in following reaction sequence which is "X"



Space for Rough Work

SECTION-III : (Maximum Marks: 18)

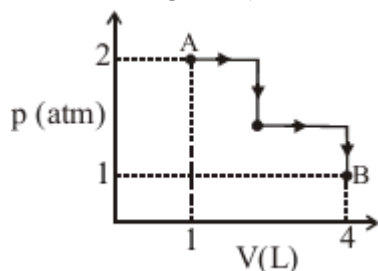
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Zero Marks : 0 If no answer is given.

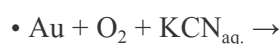
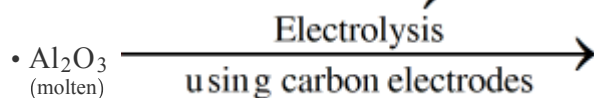
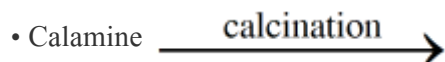
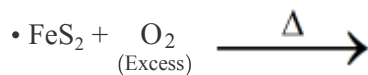
Negative Marks : -1 In all other cases

1. Calculate entropy change (in cal/K) for 1 mole of diatomic ideal gas undergoing for a process A to B as shown in diagram. ($R = 2 \text{ cal/mole-K}$; $\ln 2 = 0.7$)



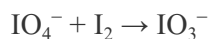
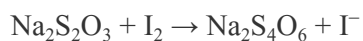
Write answer excluding decimal places.

2. Write the sum of the oxidation number of metal in products of given reactions :-



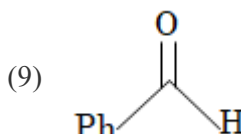
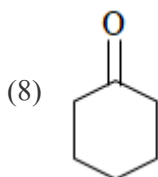
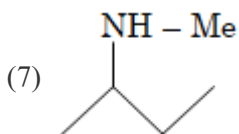
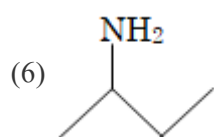
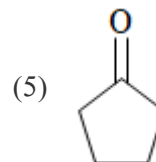
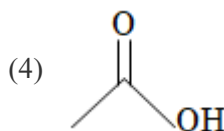
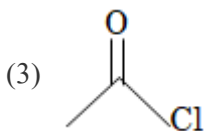
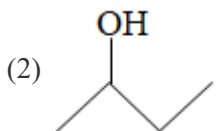
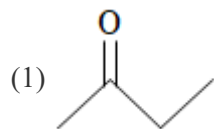
Space for Rough Work

3. 100 ml solution containing 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ & 0.2 M IO_4^- is titrated with I_2 according to following reaction.



Calculate millimoles of I_2 consumed ?

4. Identify compounds give yellow ppt with 2,4 DNP



Space for Rough Work

5. Consider the table given, which shows both the diagonal relationship between lithium and magnesium, and also the typical behaviour of lithium in group-I. We have filled for one metal(s) to give you an idea of the amount of detail you need.

Fill the blank column as (✓) if given property is similar as given in one column and (✗) if given property is different as given in one column.

Property	Li	Rest of Group I metals	Mg
Combustion with excess oxygen	O ²⁻ containing anion is main product	-----	-----
Heating with nitrogen gas	-----	-----	Nitride formation
Effect of heat on nitrates	-----	Brown gas NOT Produces	-----
Effect of heat on carbonates	-----	-----	Metal oxide with CO ₂
Heating of metal chloride in oxidizing flame	-----	Colour on flame	-----
Physical state of hydrogen carbonates	bicarbonate does not exist as a solid	-----	-----
Solubility of carbonates	Sparingly soluble/insoluble	-----	-----

Find total number of (✓) in given table after filling of blank columns.

6. Identify total number of aldehyde with molecular formula C₅H₁₀O which on reaction with “NH₂-OH” at Ph = 4 – 6 give diastereo isomeric products

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If for triangle ABC, circum radius (R), inradius(r) & ex-radius (r_1) are $\frac{13}{2}$, 2, 3 respectively, then (usual notations are used)
- (A) area of the triangle is 30 sq. units.
- (B) triangle is obtuse angled
- (C) triangle is right angled
- (D) perimeter of the triangle is 30 units.

Space for Rough Work

2. Let $f(n) = \cot^{-1} \left(\frac{9}{2} \right) + \cot^{-1} \left(\frac{33}{4} \right) + \cot^{-1} \left(\frac{129}{8} \right) + \dots$ upto n terms
- (A) $\cot \left(\lim_{n \rightarrow \infty} f(n) \right) = 2$
- (B) $(1 + 4 \cdot 2^n) \tan(f(n)) = 2 + 2^2 + 2^3 + \dots$ upto n terms
- (C) $\cot \left(\lim_{n \rightarrow \infty} f(n) \right) = \frac{1}{2}$
- (D) $(1 + 4 \cdot 2^n) \tan(f(n)) = 1 + 2 + 2^2 + 2^3 + \dots$ upto n terms
3. From any point P on the line $L : 3x + 4y = 125$ two tangents are drawn to the circle $S : x^2 + y^2 = 25$ which touches the circle at A and B .
- (A) If area of ΔPAB is minimum then length of AB is equal to $4\sqrt{6}$ units.
- (B) If area of ΔPAB is minimum then length of AB is equal to $8\sqrt{6}$ units.
- (C) If area of ΔPAB is minimum then circumradius of the ΔPAB is equal to 12.5 units.
- (D) Equation of AB can be of the form $p(4x - 3y) + q(2x + y - 2) = 0$, $p, q \in \mathbb{R}$.

Space for Rough Work

4. Let A(4, -2) and B(0, 1) are two points on a parabola. Normals at A & B to the parabola intersect at P(2, 2). Tangent parallel to AB to the parabola touches the parabola at Q and tangents drawn at A & B to the parabola meet at R.
- (A) Equation of directrix of the parabola is $2x + y + 1 = 0$
- (B) Equation of directrix of the parabola is $y = -3$
- (C) Length of latus rectum of parabola is $\frac{8}{5}$
- (D) Area of triangle AQB is equal to $\frac{5}{2}$
5. If $\sum_{n=1}^k \left(\frac{1}{n} \left(\prod_{i=1}^n \left(\sum_{r=1}^i \frac{1}{r(r+1)} \right) \right) \right) = \frac{p}{q}$; then $q - p$ divides
- (A) 1 (B) 2 (C) 3 (D) 4
6. Five fair dice are tossed. Which of the following is/are true ?
- (A) mean of number of dice showing even numbers is 2.5
- (B) variance of number of dices showing even numbers is 1.25
- (C) probability that same numbers appeared on all the 5 dice is $\frac{1}{6^4}$
- (D) probability that atleast two dice show even number, if atleast one die show even number is $\frac{26}{31}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. The contents of 3 bags w.r.t. green and red marbles is as given in the table shown.

Bag	Green	Red
A	3	1
B	2	2
C	1	3

A child randomly selects one of the bags and draws a marble from it and retains it. If the marble is green, the child draws the second marble randomly from one of the remaining bags. If the first marble drawn is red the child draws one more marble from the same bag. The probability that the second drawn marble is green is expressed as $\frac{m}{n}$ (where m & n are coprime). Find the value of $\frac{n}{m-1}$.

Space for Rough Work

2. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix}$ and $X = \frac{1}{4} \begin{bmatrix} \sqrt{6} + \sqrt{2} & -\sqrt{6} + \sqrt{2} \\ \sqrt{6} - \sqrt{2} & \sqrt{6} + \sqrt{2} \end{bmatrix}$

If $P = A X B$, $Q = B X^T A$ and $\text{tr}((PQ)^{10}) = a^{10}(1 + b^{10})$, where $a > b$ and $a, b \in \mathbb{N}$ then the value of $b + a$ is equal to (where $\text{tr}(M)$ denotes trace of matrix M & M^T denotes transpose of M).

3. Let $A = \begin{bmatrix} 4 & 1 \\ -9 & -2 \end{bmatrix}$ and $A^{100} = \begin{bmatrix} a_1 & a_3 \\ a_2 & a_4 \end{bmatrix}$ then the value of $a_1 - a_4$ is equal to

4. If $\sum_{k=0}^{10} \frac{(k-1)^2}{(k+1)(k+2)} {}^{10}C_k = \frac{a \cdot 2^{10} - b}{c}$, where $a, b, c \in \mathbb{N}$ and b is coprime with both 'a' and 'c'. The value of $\frac{11 \cdot (a+b)}{c}$ is equal to

Space for Rough Work

5. Let a function 'f' satisfies $f(-x) = f(x)$ and $f(3+x) = f(1-x) \forall x \in \mathbb{R}$ and $f(x) = \begin{cases} x & ; 0 \leq x \leq 1 \\ 1-2x & ; 1 < x \leq 2 \end{cases}$.

The number of points where function $f(x)$ is discontinuous in $[0, 100]$ is equal to D and value of $\int_0^{100} f(x) dx$ is

equal to I. The value of $2D + I$ is equal to

6. Let f be a continuous function and satisfies $f'(x) > 0$ on $(-\infty, \infty)$ and the value of $f''(x) \forall x \in (0, \infty)$ is equal to minimum value of $\min \{e^{-|x|} + 2, |x| + 2\}$. If $L = \lim_{x \rightarrow \infty} \frac{3x^2 - \frac{3}{x^2+1} - 4f'(x)}{f(x)}$ then the value of $[L^2]$ is equal to (Note : $[k]$ denotes greatest integer less than or equal to k)

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

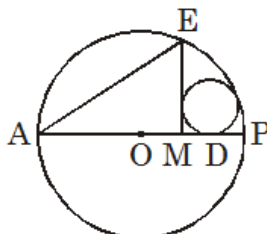
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. Let $f(x)$ be a non constant thrice differentiable function defined on 'R' such that $f(x) = f(6 - x)$ and $f'(0) = 0 = f'(2) = f'(5)$. If n is the minimum number of roots of $f''(x)^2 + f'(x)f'''(x) = 0$ in $[0, 6]$ then value of $\frac{n}{2}$ is
2. A curve passing through $(1, 1)$ and $(0, k)$ satisfies the differential equation $(y + ye^{x/y}) dx + e^{x/y} (y - x) dy = 0$, then $[k]$ is equal to (where $[.]$ denotes greatest integer function)
3. A straight line L intersects perpendicularly both the lines $\frac{x+2}{2} = \frac{y+6}{3} = \frac{z-34}{-10}$ and $\frac{x+6}{4} = \frac{y-7}{-3} = \frac{z-7}{-2}$ then the square of perpendicular distance of origin from L is equal to

Space for Rough Work

4. Let $\vec{a}, \vec{b}, \vec{c}$ be non-coplanar unit vectors equally inclined to one another at an angle of $\frac{\pi}{3}$. If $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$ then $2(p^2 + q^2 + r^2)$ is equal to (where p, q, r are scalars)
5. Let 'O' be centre of circle as shown in figure and AE be a chord from A the diameter AP is drawn and M is foot of perpendicular drawn from E on AP. A circle is drawn such that it touches EM, AP at D and circle whose diameter is AP, then the value of $\frac{AE}{AD}$ is _____



6. In a ΔABC , if a, b, c are in arithmetic progression and $\frac{b}{c} \sin 2C + \frac{c}{b} \sin 2B + \frac{b}{a} \sin 2A + \frac{a}{b} \sin 2B = 2$, then the value of $\operatorname{cosec} B$ is equal to (Note : All symbols have usual meaning for ΔABC)

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work